

Answer:

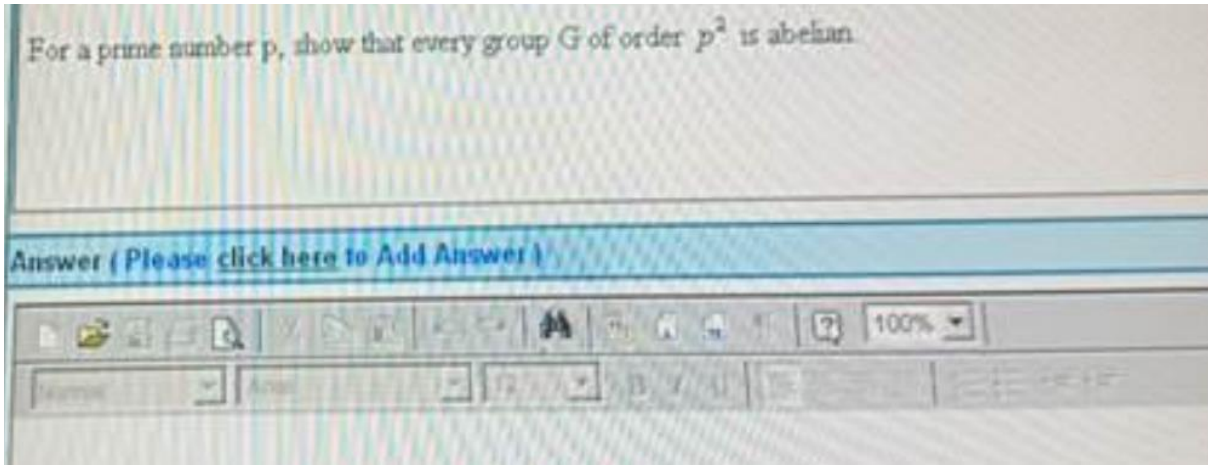
Solution

We need to show that $y(s+t)=y(s)+y(t)$ for $s, t \in \mathbb{Z}$. Using the division algorithm, we let

$s=q_1n+r_1$ (1) and

$t=q_2n+r_2$ (2) where $0 \le r_i < n$ for $i=1, 2$.

If $r_1+r_2=q_3n+r_3$ (3) for $0 \le r_3 < n$ then adding Eqs. (1) and (2) we see that $s + t = (q_1 + q_2 + q_3)n + r_3$, so that $y(s+t)=r_3$. From Eqs. (1) and (2) we see that $y(s) =r_1$ and $y(t)=r_2$. Equation (3) shows that the sum r_1+r_2 in $\mathbb{Z}_n$ is equal to r_3 also.



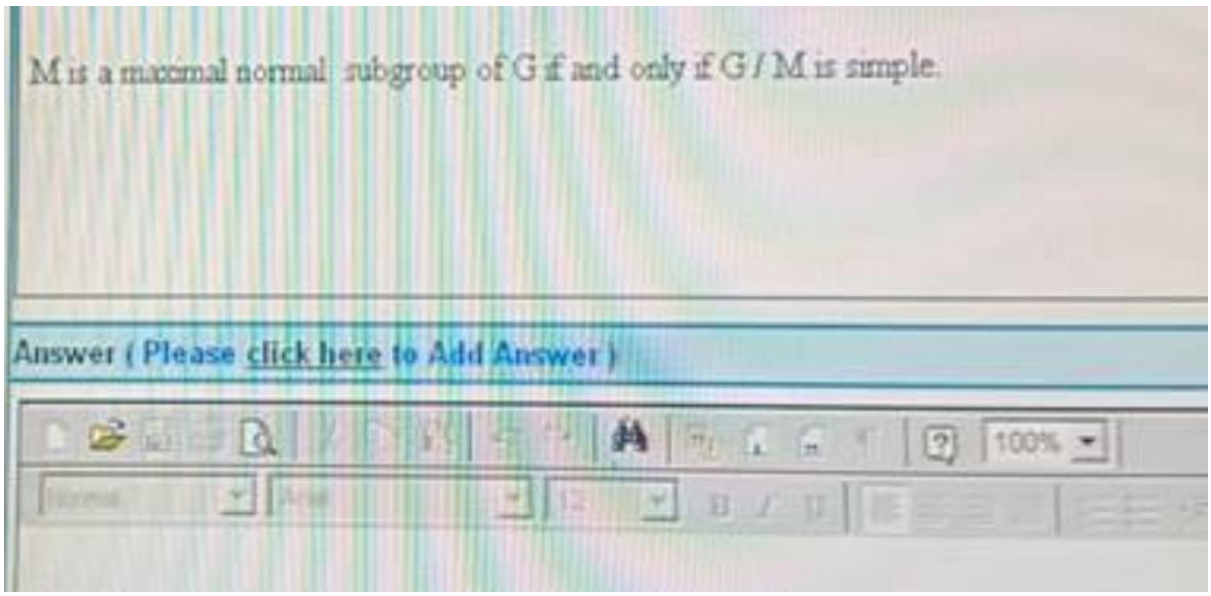
Answer:

Proof

If G is not cyclic, then every element except e must be of order p.

Let a be such an element. Then the cyclic subgroup $\langle a \rangle$ of order p does not exhaust G.

Also let $b \in G$ with $b \notin \langle a \rangle$. Then $\langle a \rangle \cap \langle b \rangle = \{e\}$, since an element c in $\langle a \rangle \cap \langle b \rangle$ with $c \neq e$ would generate both $\langle a \rangle$ and $\langle b \rangle$, giving $\langle a \rangle = \langle b \rangle$, contrary to construction.

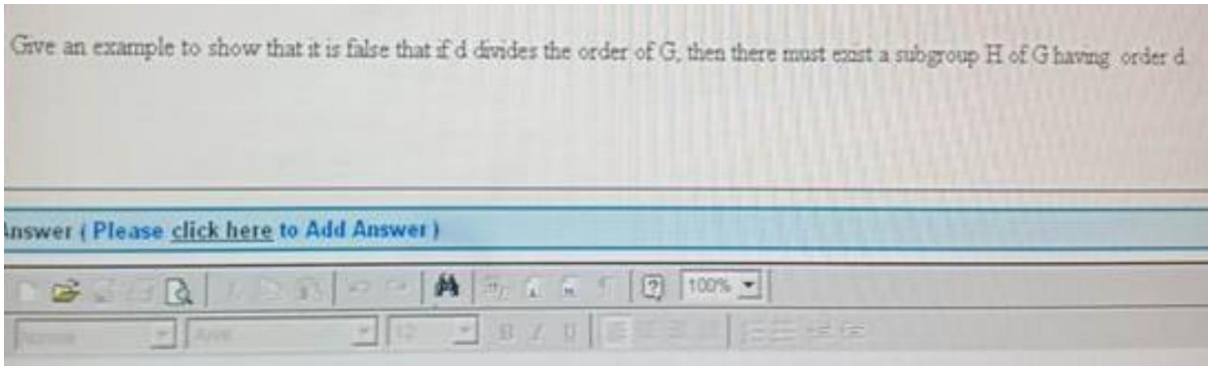


Answer:

Proof

Let M be a maximal normal subgroup of G. Consider the canonical homomorphism

$y: G \to G/M$. Now y^{-1} of any nontrivial proper normal subgroup of G/M is a proper normal subgroup of G properly containing M. But M is maximal, so this can not happen. Thus G/M is simple.

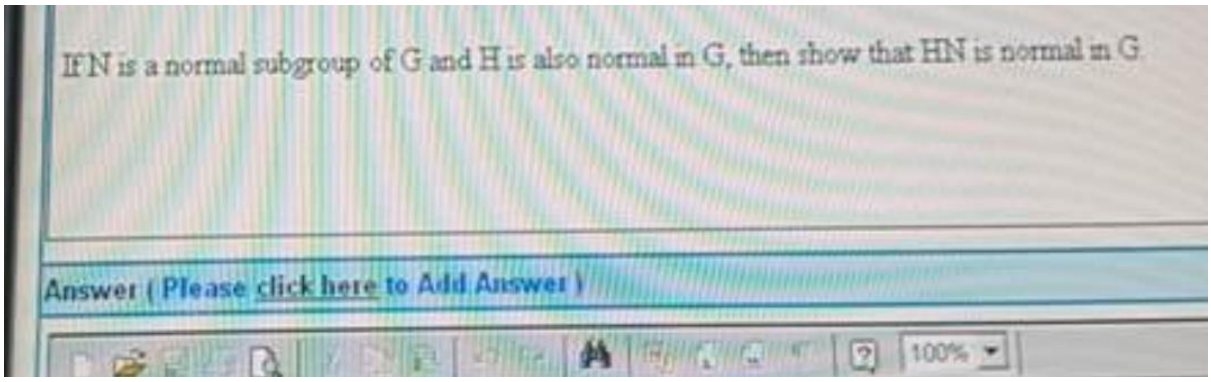


Answer:

Example

We show that A_4, which has order 12, contains no subgroup of order 6.

Suppose that H were a subgroup of A_4 having order 6.



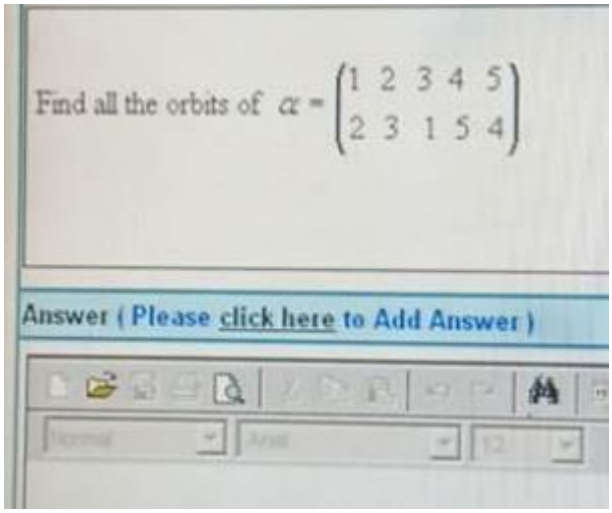
Answer:

Lemma

If N is a normal subgroup of G, and if H is any subgroup of G, then

H V N=HN=NH.

Furthermore, if H is also normal in G, then HN is normal in G.



Answer:

- Look at what happens to elements as a permutation is applied.
- $$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 1 & 5 & 4 \end{pmatrix}$$

$\alpha(1)=2, \alpha^2(1)=3, \alpha^3(1)=1$ {1,2,3}

$\alpha(4)=5, \alpha^2(4)=4$ {4,5}



ANSWER:

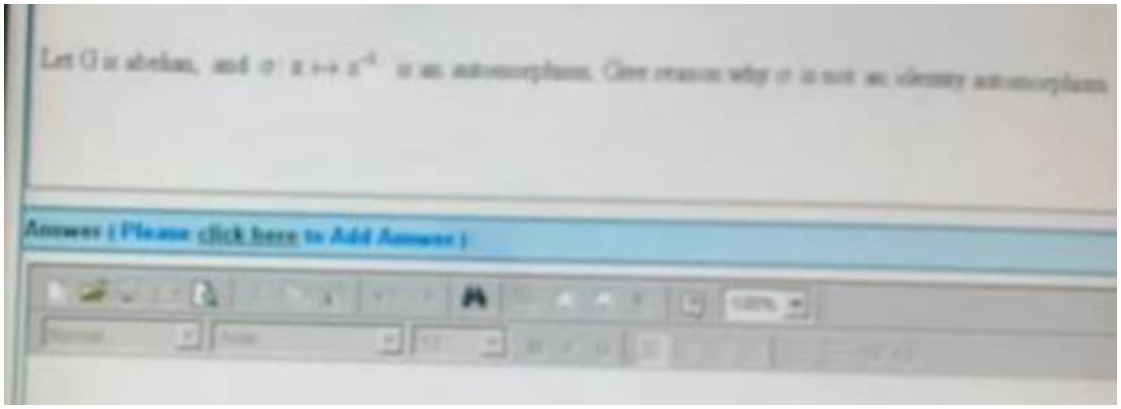
Proof

If N is a normal subgroup of G and G/N is abelian, then

$(a^{-1}N)(b^{-1}N)=(b^{-1}N)(a^{-1}N)$;
that is, $aba^{-1}b^{-1}N=N$,

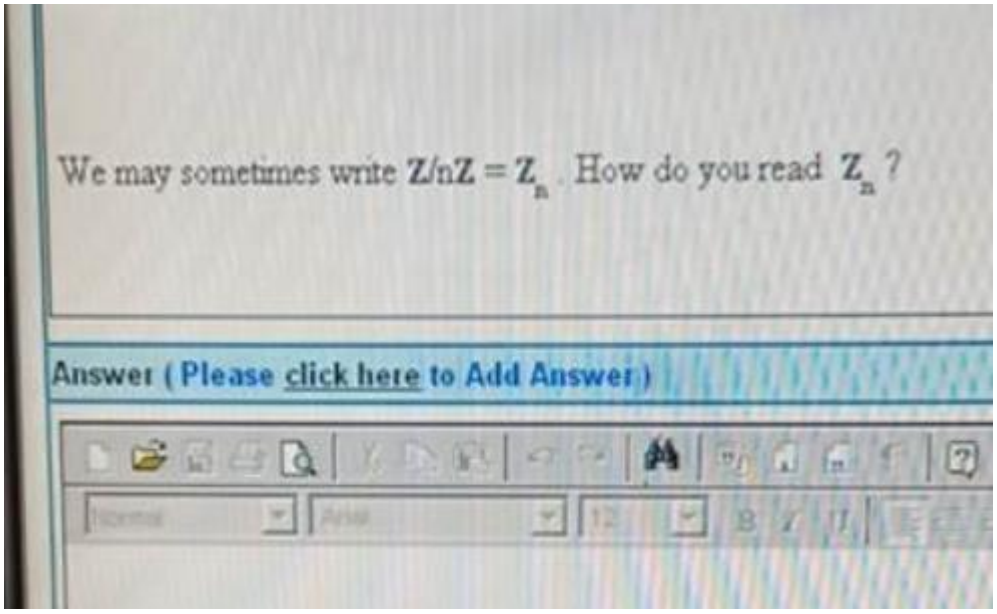
so $aba^{-1}b^{-1} \in N$, and

$C \leq N$.



Answer:

When G is abelian, then $\sigma: x \mapsto x^{-1}$ is an automorphism, and, clearly, σ is not an identity automorphism.
When G is not abelian, there exists a nontrivial inner automorphism.



Answer:

Theorem, we see that the factor group $\mathbb{Z}/n\mathbb{Z}$ is isomorphic to $\mathbb{Z}_n$. The cosets of $n\mathbb{Z}$ are the residue classes modulo n.